

Daily Progress Report

Week 3_Day4: Forms & Input Validation Fundamentals

Program	3-Month MERN Stack Internship Program
Intern Name	Mohsin Ali
Week	Week_3 (Day 4)
Date	9 July, 2026

Day 4: Forms & Input Validation Fundamentals

1.1 Topics Studied

- Preventing default form submission using `e.preventDefault()`
- Reading form data using the `FormData` object
- Writing custom validation logic in JavaScript
- Displaying inline error messages using DOM manipulation
- Building responsive and user-friendly forms

1.2 Task Description

Developed a responsive **User Registration Form** that performs client-side validation using JavaScript. The form collects user information and validates the password before allowing submission. The validation ensures that the password is at least 8 characters long and matches the **Confirm Password** field. Instead of relying on the browser's default validation, custom inline error messages are displayed to provide clear feedback to the user.

1.3 Implementation Details

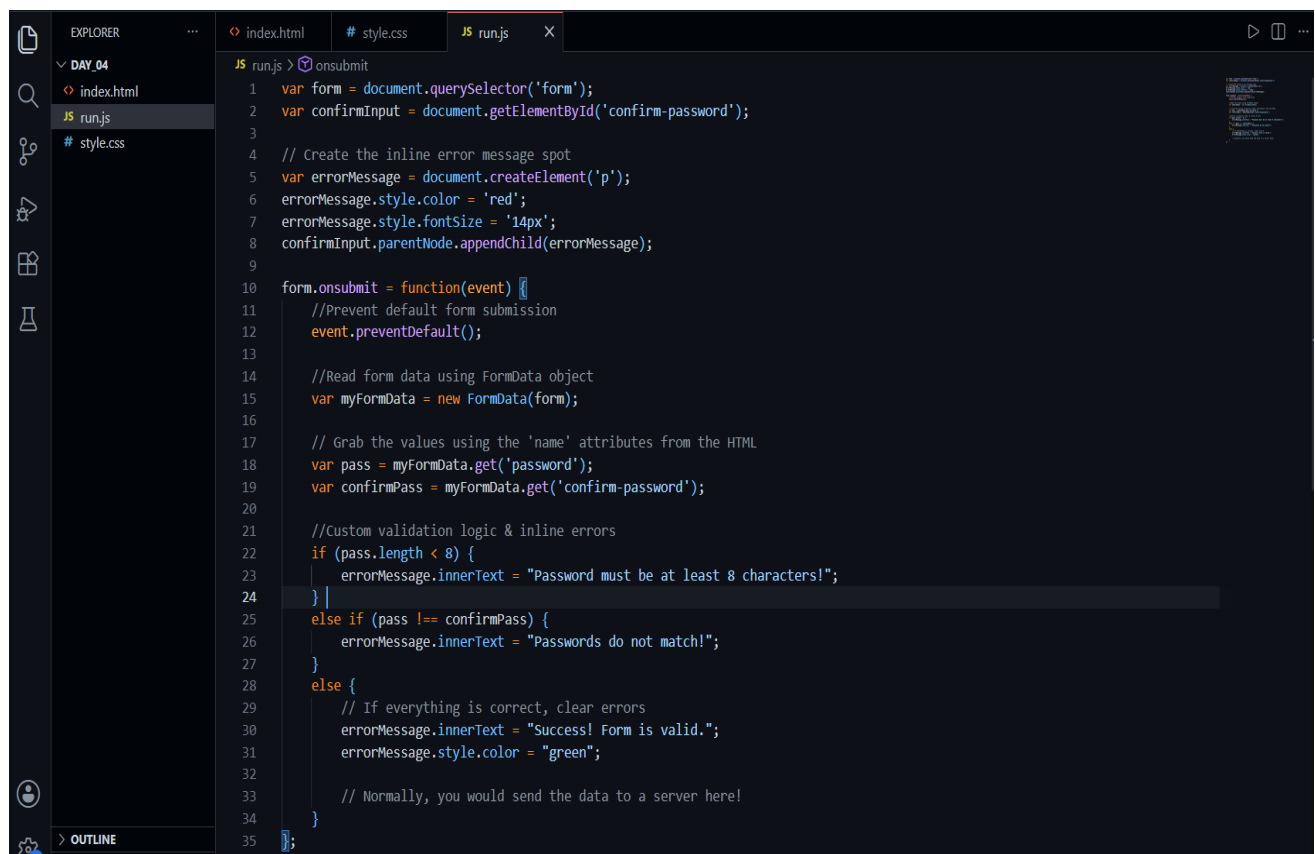
- Designed and styled a responsive registration form using HTML and CSS with a clean and user-friendly interface.
- Handled the form submission using JavaScript and prevented the default submission behavior with `e.preventDefault()`.
- Used the `FormData` object to retrieve user input values from the form efficiently.
- Implemented custom validation logic to verify that the password contains at least 8 characters.
- Added a second validation to ensure that the **Password** and **Confirm Password** fields contain the same value before submission.

- Displayed meaningful inline error messages whenever validation failed, helping users identify and correct input errors.
- Ensured the form remained responsive and easy to use across different screen sizes.

1.4 Outcome

The User Registration Form was successfully developed with responsive design and custom client-side validation. This task enhanced my understanding of JavaScript form handling, the FormData API, and validation techniques. It also demonstrated how custom validation and inline feedback can improve usability and provide a better overall user experience compared to default browser validation.

Screenshot:



The screenshot shows a code editor with three tabs: index.html, style.css, and run.js. The run.js tab is active, displaying JavaScript code for form validation. The code includes comments and logic for preventing default form submission, reading form data using the FormData API, and implementing custom validation logic for password length and matching. The error message is dynamically created and styled.

```
JS run.js > onsubmit
1  var form = document.querySelector('form');
2  var confirmInput = document.getElementById('confirm-password');
3
4  // Create the inline error message spot
5  var errorMessage = document.createElement('p');
6  errorMessage.style.color = 'red';
7  errorMessage.style.fontSize = '14px';
8  confirmInput.parentNode.appendChild(errorMessage);
9
10 form.onsubmit = function(event) {
11     //Prevent default form submission
12     event.preventDefault();
13
14     //Read form data using FormData object
15     var myFormData = new FormData(form);
16
17     // Grab the values using the 'name' attributes from the HTML
18     var pass = myFormData.get('password');
19     var confirmPass = myFormData.get('confirm-password');
20
21     //Custom validation logic & inline errors
22     if (pass.length < 8) {
23         errorMessage.innerText = "Password must be at least 8 characters!";
24     }
25     else if (pass !== confirmPass) {
26         errorMessage.innerText = "Passwords do not match!";
27     }
28     else {
29         // If everything is correct, clear errors
30         errorMessage.innerText = "Success! Form is valid.";
31         errorMessage.style.color = "green";
32
33         // Normally, you would send the data to a server here!
34     }
35 };
```

Screenshots:

Registration

Full Name: Mohsin Ali

Username: Rana

Email: mohsinalirana222@gmail.com

Phone Number: 0300088299

Password:

Confirm Password:

Gender: ☒ Male ☐ Female ☐ Prefer not to say

Register

Password must be at least 8 characters!

Registration

Full Name: Mohsin Ali

Username: Rana

Email: mohsinalirana222@gmail.com

Phone Number: 0300088299

Password:

Confirm Password:

Gender: ☒ Male ☐ Female ☐ Prefer not to say

Register

Passwords do not match!

Registration

Full Name

Mohsin Ali

Username

Rana

Email

mohsinalirana222@gmail.com

Phone Number

03000882778

Password

.....

Confirm Password

.....

Success! Form is valid.

Gender

☒ Male ☐ Female ☐ Prefer not to say

Register